

Routing Information Protocol (RIP)

The Routing Information Protocol ('RIP') is one of the oldest distance-vector routing protocols which employ the hop count as a routing metric. RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from source to destination. The largest number of hops allowed for RIP is 15, which limits the size of networks that RIP can support. In RIPv1 routers broadcast updates with their routing table every 30 seconds.

- **RIP version 1**

The original specification of RIP, defined in was published in 1988. When starting up, and every 30 seconds thereafter, a router with RIPv1 implementation broadcasts to 255.255.255.255 a request message through every RIPv1 enabled interface. Neighbouring routers receiving the request message respond with a RIPv1 segment, containing their routing table. The requesting router updates its own routing table, with the reachable IP network address, hop count and next hop, that is the router interface IP address from which the RIPv1 response was sent. As the requesting router receives updates from different neighbouring routers it will only update the reachable networks in its routing table, if it receives information about a reachable network it has not yet in its routing table or information that a network it has in its routing table is reachable with a lower hop count. Therefore, a RIPv1 router will in most cases only have one entry for a reachable network, the one with the lowest hop count. If a router receives information from two different neighbouring router that the same network is reachable with the same hop count but via two different routes, the network will be entered into the routing table two times with different next hop routers. The RIPv1 enabled router will then perform what is known as equal-cost load balancing for IP packets.

RIPv1 enabled routers not only request the routing tables of other routers every 30 seconds, they also listen to incoming requests from neighbouring routers and send their own routing table in turn. RIPv1 routing tables are therefore updated every 25 to 35 seconds.

- **RIP version 2**

Due to the deficiencies of the original RIP specification, RIP version 2 (RIPv2) was developed in 1993 and proclaimed Internet Standard 56 in 1998. It included the ability to carry subnet information, thus supporting Classless Inter-Domain Routing (CIDR). To maintain backward compatibility, the hop count limit of 15 remained.

RIPv2 multicasts the entire routing table to all adjacent routers at the address 224.0.0.9, as opposed to RIPv1 which uses broadcast.